# WEEKLY PROGRESS REPORT

Rice LeafDisease Recognition andClassification

Reproduction Study: Vision Transformer (ViT-Small) From Scratch

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| Student | Srijita Nag |
| Programme | B.Tech – Artificial Intelligence |
| Academic Year | 2026–2027 |
| Project Stage | Week 2 Progress |
| Reporting Date | 26 October 2026 |
| Scope of this report | Standalone ViT architecture from scratch → training → comparison with the research paper |

## 1. Executive Summary

This week's work continued from the previous report, which documented the dataset preparation and EfficientNet-B0 CNN architecture implemented from scratch. The current stage focused on building a standalone Vision Transformer (ViT-Small) from scratch, verifying its main components, and completing an eight-epoch training run. The implementation includes patch embedding, positional embeddings, transformer encoder blocks with multi-head self-attention and feed-forward layers, and a four-class classification head.

The best observed test accuracy was 78.50% at Epoch 3; the final Epoch 8 test accuracy was 78.00%. The result is recorded as a baseline for the next stage. The original research paper uses ImageNet-pretrained EfficientNet-B0 and ViT-Small, while our project intentionally trains both architectures from scratch. Consequently, the paper's scoresare reference benchmarks rather than directly equivalent results.

## 2. Short Summary of the Actual Research Paper

Paper:“Efficient-ViT B0Net: A high-performance light weight transformer for rice leaf disease recognition and classification” by Santosh Kumar Upadhyaya and Rajesh Prasad, Journal of the Nigerian Society of Physical Sciences, Volume 7 (2025), Article 2940.

The paper proposes a hybrid CNN–Vision Transformer model for four-class rice-leaf classification: Bacterial Leaf Blight, Brown Spot, Health, and Leaf Smut. EfficientNet-B0 extracts compact local and fine-grained texture features, while ViT-Small models global relationships between image features. The paper uses pretrained EfficientNet-B0 and ViT-Small (ImageNet initialization), with 224×224 images. The CNN feature map of 7×7×320 is flattened into 49 tokens and projected from 320 to 384 dimensions before entering the ViT component.

The dataset is described as 16,000 images, with 4,000 images per class and an 80:20 split (12,800 training and 3,200 testing images). The paper reports 91.3% for EfficientNet-B0, 97.0% for standalone ViT-Small, and 99.13% for the proposed Efficient-ViT B0Net hybrid. The paper also reports 99.13% macro precision, recall, and F1-score for the hybrid model.

## 3. Critical Methodological Difference: Paper vs. Our Project

| Component | Original research paper | Our project |
|---|---|---|
| EfficientNet-B0 | Pretrained CNN architecture | Implemented and trained from scratch; previous report recorded 99.47% best test accuracy |
| ViT-Small | Pretrained on ImageNet | Implemented and trained from scratch; best test accuracy 78.50% |
| ViT role | Standalone benchmark and part of proposed hybrid | Standalone baseline built first; hybrid remains future work |
| Main purpose | Evaluate pretrained hybrid pipeline | Study and train the architecture without pretrained weights |

Why this distinction matters: pretrained weights already encode features learned from a large external dataset. A scratch model starts with randomly initialized weights and must learn its representations from the project data. The performance gap is therefore an important experimental observation, not by itself proof of an incorrect implementation.

# 4. Project Objective and Workflow

The objective is to understand and progressively implement the selected paper's architecture. The planned workflow is: dataset and preprocessing → EfficientNet-B0 CNN from scratch → standalone ViT-Small from scratch → scratch EfficientNet-B0 + ViT-Small hybrid → comparison with the paper's published results. This report documents thestandalone ViT stage; thehybrid model hasnotyet been trained.

# 5. Work Completed This Week

## ViT configuration

Defined the ViT-Small architecture settings for scratch training.

### Patch embedding

Implemented image-to-patch/token embedding stage.

### Positional embedding

Implemented positional information for the token sequence.

### Transformer encoder

Implemented multi-head self-attention, residual connections, normalization, and feed-forward MLP blocks.

### Encoder stack

Constructed 12 transformer blocks.

### Classification head

Added a four-class output head for rice-leaf disease classification.

### Parameter verification

Notebook reported 21,666,436 trainable parameters (approximately 21.67 million).

### Training

Completed an eight-epoch training run and recorded train/test loss and accuracy.

# 6. ViT-Small Architecture Implemented From Scratch

| Component | Implementation / role |
|---|---|
| Input image | RGB image, 224×224×3 |
| Patch embedding | Splits image into patches and maps patches to embedding vectors |
| Positional embedding | Adds information about token positions |
| Transformer encoder | 12 blocks using multi-head self-attention and feed-forward MLP |
| Normalization and residuals | Support stable information flow through encoder blocks |
| Classification head | Maps the final representation to 4 rice-leaf classes |
| Trainable parameters | 21,666,436 (approximately 21.67M) |

In contrast to the paper's hybrid pathway, the current ViT experiment takes the image input into the standalone ViT architecture. The paper's 7×7×320 → 49×320 → 49×384 interface is relevant to the future hybrid implementation and is not claimedas part of this standalone ViT run.

# 7. ViT Training Configuration

| Parameter | Value |
|---|---|
| Model | Standalone ViT-Small from scratch |
| Input | 224×224 RGB images (project pipeline) |
| Epochs | 8 |
| Optimizer / loss | Use the exact settings from the Colab notebook; prior CNN setup used AdamW and Cross-Entropy Loss |
| Execution | Google Colab GPU (project environment) |

Note: Only the eight-epoch duration and metrics are treated as confirmed here. Confirm the precise optimizer, learning rate, batch size, and input pipeline from the current ViT notebook before final submission if these are required by the supervisor.

## 8. ViT-Small Scratch Training Results

The following values are transcribed from the current ViT-Small training output. Accuracy is expressed as a percentage; loss is cross-entropy loss.

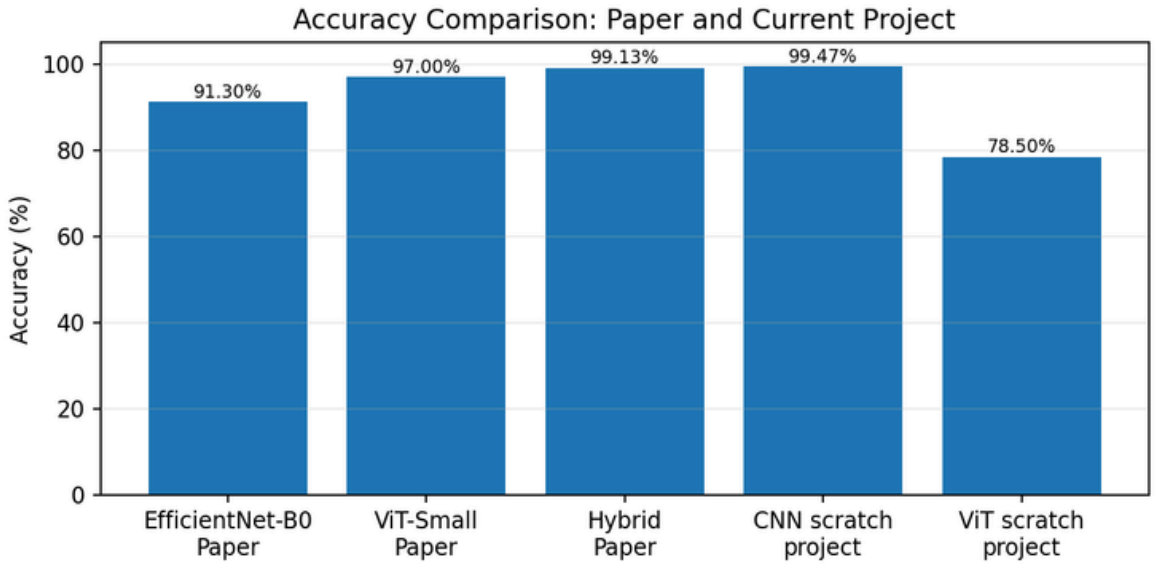
| Epoch | Train Loss | Train Acc. | Test Loss | Test Acc. |
|---|---|---|---|---|
| 1 | 1.3149 | 36.05% | 1.4415 | 38.84% |
| 2 | 1.0392 | 52.82% | 0.8032 | 65.44% |
| 3 | 0.6110 | 74.21% | 0.5070 | 78.50% |
| 4 | 0.7233 | 69.23% | 0.9760 | 57.38% |
| 5 | 0.7674 | 66.84% | 0.6892 | 74.38% |
| 6 | 0.6207 | 74.70% | 0.7293 | 71.31% |
| 7 | 0.6994 | 70.42% | 1.0505 | 64.41% |
| 8 | 0.8110 | 65.99% | 0.5159 | 78.00% |

Best scratch ViT result:78.50% test accuracy at Epoch 3. Final Epoch 8 test accuracy: 78.00%.

## 9. Comparison With the Main Research Paper Results

This comparison distinguishes thepaper's reported pretrained-model benchmarks fromourscratch-training results. These are not fully controlled comparisons unless dataset identity, splits, preprocessing, augmentation, initialization, and training configuration are confirmed to match.

| Model / result | Research paper | Our project | Interpretation |
|---|---|---|---|
| EfficientNet-B0 | 91.3% | 99.47% best test accuracy | Scratch CNN baseline from previous report; setup differs. |
| ViT-Small | 97.0% | 78.50% best test accuracy | Paper uses ImageNet-pretrained ViT; project ViT trained from scratch. |
| Proposed Efficient-ViT B0Net hybrid | 99.13% | Not yet implemented | Hybrid is the next stage. |




The scratch ViT's best observed test accuracy is 18.50 percentage points below the paper's standalone ViT-Small result. This is an important baseline observation, but it should be interpreted alongside the use of ImageNet pretraining in the paper. The previous scratch CNN result of 99.47% is higher than the paper's EfficientNet-B0 result of91.3%; differing experimental setups mean that this alone does not establish superiority.

## 10. Discussion

ThestandaloneViT-Smallarchitecture has been constructed and trained, so the project has completed the planned step after the CNN baseline. The run reached 78.50% test accuracy at Epoch 3, but later results fluctuated. A next experiment should review learning rate, regularization, augmentation, and validation-based checkpoint selection. Precision, recall, F1-score, and a confusion matrix should be added to understand class-wise performance.

The selected paper's main contribution is the combination of local CNN features with global transformer attention. The next technical milestone is to connect the scratch CNN's 7×7×320 feature map to 49×320 tokens and project theseto 49×384beforethe ViT encoder. The hybridarchitecture has not yet been evaluated in our project.

## 11. Skills and Concepts Learned

- VisionTransformerarchitectureandpatch/tokenembedding
- Positional embeddings and multi-head self-attention
- Transformer encoder blocks, residual connections, normalization, and MLP
- Building a classification model from scratch without pretrained weights
- Tracking epoch-wise training and test metrics
- Interpreting accuracy and cross-entropy loss curves
- Comparing scratch-training resultswith pretrainedresearch benchmarks

## 12. Next Week's Planned Work

• Review the standalone ViT training stability and document final evaluation metrics.

• Calculate precision, recall, F1-score, and the confusion matrix for the scratch ViT.

• Implement the hybrid interface: 7×7×320 → 49×320 → 49×384.

• Combine the scratch EfficientNet-B0 and scratch ViT-Small architectures.

• Train and evaluate the scratch hybrid using a documented, consistent data split.

• Compare the final scratch hybrid result with the paper's reported 99.13% hybrid result, noting the pretrained-versus-scratch distinction.

## 13. References

[1] Upadhyaya, S. K., & Prasad, R. (2025). “Efficient-ViT B0Net: A high-performance lightweight transformer for rice leaf disease recognition and classification.” Journal of the Nigerian Society of Physical Sciences, 7, Article 2940. DOI: 10.46481/jnsps.2025.2940.

[2] Dosovitskiy, A., et al. (2021). “An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale.” International Conference on Learning Representations (ICLR). https://arxiv.org/abs/2010.11929

[3] Project implementation: ViT-Small from-scratch architecture and eight-epoch training output from the student's Google Colab notebook, October 2026.

Reporting note: The dataset details summarized from the paper are not independently revalidated against every file in the current notebook here. Confirm that the project's actual split and training configuration match the statements before final submission.